

GOVERNMENT SCHOOL MANAGEMENT SYSTEM

Role-Based Access & Functionality Report

What Each Role Can and Cannot Do, Illustrated Against a Reference School

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1. Purpose

This report defines, role by role, exactly what a GSMS user can and cannot do, and why. The SRS (GSMS-SRS-001) states that access is role-based; this document makes that concrete by reasoning through a single, consistent reference school so every restriction can be checked against a real situation rather than an abstract rule. It is intended for the development team (as the source of truth for RBAC rules), for QA (as the basis for permission test cases), and for school leadership reviewing the system before rollout.

2. Reference School Used Throughout This Report

To keep every example concrete, this report uses one consistent, fictional government school: **Green Valley Government School, Trincomalee** — Grades 1 to 13, approximately 1,450 students across 42 classes (sections A–E per grade where enrolment requires it), 68 teaching staff and 14 non-teaching staff. The people below recur through the rest of this report:

Name	Role
Mrs. Perera	Principal
Mr. Rathnayake	Vice Principal
Ms. Ilangakoon	School Administrator
Mr. Jayasinghe	Class Teacher, Grade 10A, and Mathematics subject teacher
Mrs. Kumari	Administrative Staff (records & welfare desk)
Ms. Rajaratnam	Librarian
Mr. Wickrama	Laboratory Assistant (Science lab)
Kavindi Silva	Student, Grade 10A
Mrs. Silva	Parent of Kavindi Silva
Mr. Bandara	Zonal Education Officer, Trincomalee Zone

Every “can/cannot” statement in Sections 4–13 is checked against how these specific people would use the system on an ordinary school day, so the logic stays grounded rather than hypothetical.

3. Access Control Principles

- **Least privilege** — a role gets exactly the access its real-world job requires, no more. A Librarian never needs to see examination marks, so the system never shows them.
- **Scope before action** — before asking “can this role do X,” GSMS first asks “which records can this role do X to.” A Teacher can mark attendance, but only for classes they are assigned to teach — not the whole school.
- **Separation of entry and approval** — whoever enters data (e.g. Mr. Jayasinghe entering marks) is generally not the same role that gives it final, visible effect (Mrs. Perera or Mr. Rathnayake approving report card publication). This mirrors how the paper process already works.
- **Guardianship, not ownership, for Parents and Students** — a Parent account can see everything relevant to their own child, and nothing about any other child, even a child in the same class.
- **Aggregation over identification for external oversight** — the Education Officer role sees patterns across schools (attendance rates, pass rates) but not individual student records, consistent with data-protection expectations for minors' data noted in the SRS.

4. Master Permission Matrix

F = Full control (create, edit, delete) · E = Enter/Edit (create/update, no delete) · A = Approve/publish only
 · O = Own-scope only (limited to own class/child/records) · R = Read-only · – = No access.

Module	Prin.	VP	Admin	Teacher	A.Staff	Libr.	Lab	Stud.	Parent	EdOf f.
Student Mgmt	F	F	F	O	E	R	-	O	O	-
Teacher Mgmt	F	F	F	O	-	-	-	-	-	-
Staff Mgmt	F	F	F	-	O	-	-	-	-	-
Academic Mgmt	F	F	F	O	-	-	-	R	R	-
Attendance	R	R	E	O	E	-	-	O	O	R
Timetable	A	A	F	R	-	-	-	O	O	-
Examination	A	A	E	O	-	-	-	O	O	R
Library	R	R	R	-	-	F	-	O	O	-
Inventory	R	R	F	-	E	O	O	-	-	-
Welfare	A	A	F	R	E	-	-	O	O	R
Communication	F	F	E	E	E	-	-	R	R	-
Reports	F	F	F	O	O	-	-	-	-	R
User Mgmt	F	E	E	-	-	-	-	-	-	-

Read this row by row: e.g. in **Examination**, Mr. Jayasinghe (Teacher) has O — he can enter and edit marks only for the subjects and classes he teaches; Mrs. Perera and Mr. Rathnayake have A — they cannot type in a mark themselves, but must approve the term's results before Kavindi and Mrs. Silva can see them.

5. Principal (Mrs. Perera)

The Principal holds full authority over the school's use of GSMS but, consistent with the separation-of-entry-and-approval principle, exercises it mostly through approval and oversight rather than day-to-day data entry.

Can do

- View any student, teacher, staff or class record in the school.
- Approve and publish finalized timetables and term examination results.
- Approve welfare program enrolments and disbursements above the Administrator's own approval limit.
- Create and manage all user accounts in the school, including granting or revoking Administrator access.
- Broadcast school-wide announcements and view all cross-module reports.
- Override a restriction in an exceptional, logged case (e.g. correcting a published mark after a genuine clerical error), with the override recorded in the audit log.

Cannot do

- Directly edit day-to-day operational data such as marking attendance or entering exam marks — this stays with Teachers and Administrative Staff to keep the record of who actually did the work accurate.
- Create a user account with a role higher than Principal (there is none) or grant herself a permission outside the Principal role definition without an audit-logged override.

Worked example

When Mr. Jayasinghe submits Grade 10A's Mathematics term marks, the results sit in "pending approval" status. Mrs. Perera reviews the class average and any flagged anomalies, then approves publication. Only after her approval do Kavindi and Mrs. Silva see the marks in their accounts.

6. Vice Principal (Mr. Rathnayake)

The Vice Principal mirrors most of the Principal's approval authority to allow the school to keep functioning when the Principal is unavailable, but does not hold every Principal-only power.

Can do

- Approve timetables, examination results and welfare disbursements, functioning as a second approver alongside the Principal.
- View all student, teacher and staff records school-wide.
- Create teacher and staff user accounts.

Cannot do

- Grant or revoke School Administrator access — this remains a Principal-only action since it changes who can configure the school's core setup.
- Delete a student or teacher record outright — de-registration/transfer still requires Principal sign-off given its permanence.

Worked example

While Mrs. Perera is at a zonal meeting, a timetable conflict is found for Grade 8. Mr. Rathnayake can approve the corrected timetable himself so Grade 8 is not left without a published schedule, but if a teacher needs to be formally transferred out of the school, that waits for the Principal.

7. School Administrator (Ms. Ilangakoon)

The Administrator is GSMS's day-to-day operator: the role with the broadest hands-on data entry authority, but without final publishing or approval power over academic outcomes.

Can do

- Create, edit and de-activate Student, Teacher and Staff records, including bulk import at the start of the academic year.
- Build and edit timetables, and schedule examinations.
- Manage inventory and welfare programs end-to-end, including disbursement within a pre-set threshold.
- Create Teacher, Staff, Student and Parent user accounts, and reset passwords.

Cannot do

- Publish a finalized timetable or examination result to students and parents — that step needs Principal or Vice Principal approval.
- Grant another account School Administrator or Principal-level access.
- Enter examination marks on behalf of a subject teacher — marks entry is tied to the teacher of record for accountability.

Worked example

At the start of the year, Ms. Ilangakoon bulk-imports the new Grade 1 intake and builds the draft timetable for all 42 classes. She cannot make that timetable visible to teachers and parents until Mrs. Perera or Mr. Rathnayake approves it.

8. Teacher (Mr. Jayasinghe)

A Teacher's access is scoped to the classes and subjects actually assigned to them — the system distinguishes a Class Teacher's slightly wider view of their own class from a Subject Teacher's narrower, subject-only view.

Can do

- Mark attendance for classes they teach (Mr. Jayasinghe: Grade 10A as class teacher, plus any other class he teaches Mathematics in).
- Enter and edit examination marks for the subjects/classes assigned to him, up until the entry window closes.
- View the timetable, and send announcements/messages to the parents and students of his own classes.
- As Class Teacher of 10A specifically, view a consolidated view of 10A's attendance and welfare status across all subjects, to support pastoral care.

Cannot do

- View or edit records for a class he does not teach — Mr. Jayasinghe cannot open Grade 6C's attendance or marks.
- Edit a student's guardian contact details, enrolment status or transfer records — these are Administrator functions to keep master data consistent.
- Access the Library, Inventory or Welfare administration screens, or see another teacher's private notes/marks-in-progress before submission.
- Publish his own marks directly to students and parents without approval.

Worked example

Mr. Jayasinghe can see Kavindi's Mathematics performance and her overall 10A attendance record as her class teacher, but if Kavindi were in Grade 11B instead, he would have no access to her record at all unless he also taught a subject in that class.

9. Administrative Staff (Mrs. Kumari)

Administrative Staff handle operational record-keeping — particularly welfare and general records support — without the Administrator's full configuration authority.

Can do

- Enter and update welfare program enrolment and disbursement records (uniforms, textbooks, meal programs) under the Administrator's configured programs.
- Assist with student and staff record data entry (e.g. updating an address) without approval rights over transfers or de-registration.
- Mark staff attendance and process inventory receipts/issues for general school stock.

Cannot do

- Create or configure a new welfare program — only the Administrator defines eligibility rules and budgets; Mrs. Kumari operates within them.
- View examination marks, teacher performance data, or academic reports.
- Create user accounts or reset another user's password.

Worked example

When Kavindi's family applies for the textbook assistance program, Mrs. Kumari checks the eligibility checklist Ms. Ilangakoon configured, enrolls Kavindi, and records the textbook set as disbursed. She could not have created that textbook program herself.

10. Librarian (Ms. Rajaratnam)

The Librarian's access is deliberately narrow: full control of the library catalogue and circulation, and nothing outside it.

Can do

- Catalogue, edit and retire library items; process issue and return transactions; view overdue items across the school.
- Look up a student or teacher's name and class/section to confirm identity when issuing a book (read-only lookup, not their academic record).

Cannot do

- View a student's attendance, marks, welfare status or guardian contact details.
- Access Inventory (non-library assets), Examination, Welfare or Reports modules.

Worked example

When Kavindi borrows a reference book, Ms. Rajaratnam can see "Kavindi Silva, Grade 10A" to confirm she is a currently enrolled student, but cannot see Kavindi's attendance record or exam results from that same screen.

11. Laboratory Assistant (Mr. Wickrama)

Scoped even more narrowly than the Librarian: responsibility for laboratory equipment and consumables only.

Can do

- Record receipt, issue and condition of science lab equipment and consumables.
- Flag low-stock lab items for the Administrator's attention.

Cannot do

- Access any student, teacher or staff record.
- View or edit general school inventory outside the science lab, or any academic/attendance/examination data.

Worked example

Mr. Wickrama logs that a set of laboratory glassware was issued to the Grade 10 Science class for an experiment; he has no visibility into who is enrolled in that class beyond what the Science teacher tells him directly.

12. Student (Kavindi Silva)

A Student account is read-only and strictly limited to Kavindi's own academic life — it is a window onto her own records, not an editing tool.

Can do

- View her own timetable, attendance record, approved examination results and report cards.
- View library items currently borrowed and their due dates, and receive announcements addressed to her class or year.

Cannot do

- Edit any of her own records — attendance, marks and enrolment data are all entered and corrected by staff, never by the student.
- View any other student's data, a teacher's private notes, or any administrative/welfare/inventory information.
- See examination results before the Principal or Vice Principal has approved them.

Worked example

Kavindi can check her own Mathematics term mark the moment Mrs. Perera approves the term's results, but she cannot see Grade 10A's class average breakdown by student, and she cannot see her mark at all while it is still sitting in Mr. Jayasinghe's pending-approval queue.

13. Parent (Mrs. Silva)

A Parent account is scoped strictly to the guardian's own linked children, mirroring how a parent would only ever receive their own child's report card and absence call, never another family's.

Can do

- View attendance, approved examination results, timetable, and welfare enrolment/disbursement status for each of her own linked children.
- Receive announcements and messages addressed to her child's class or year, and reply where the Communication module allows.
- If she has more than one child at the school, switch between their linked profiles within the same account.

Cannot do

- View any information about a child who is not hers, even a classmate seated next to her own child.
- Edit attendance, marks, enrolment or welfare records — a Parent account has no write access to academic or administrative data.
- Access Library, Inventory, Reports or User Management screens.

Worked example

Mrs. Silva receives an automatic notification the day Kavindi is marked absent, and can see Kavindi's term report card as soon as it is published. If Kavindi's classmate is also marked absent that day, Mrs. Silva is never shown that fact in any form.

14. Education Officer (Mr. Bandara)

The zonal Education Officer role is designed for oversight across many schools, not case-by-case student administration, so it trades individual detail for read-only, aggregated visibility.

Can do

- View aggregated attendance rates, examination pass rates and welfare program coverage across all schools in the Trincomalee zone, including Green Valley Government School.
- Export zonal-level reports for Ministry reporting purposes.

Cannot do

- Open an individual student's, teacher's or staff member's record by name.
- Edit any data in any school's GSMS instance — the role is strictly read-only, and strictly aggregate.
- Access Library, Inventory, Communication or User Management for any school.

Worked example

Mr. Bandara can see that Green Valley Government School's Grade 10 attendance rate this term is 96%, one of the strongest in the zone, but he cannot open the roster to see that the figure includes Kavindi Silva specifically — the report never surfaces her name.

15. Edge Cases and Escalation Scenarios

Scenario	How GSMS Resolves It
A staff member is both a Teacher and a Parent at the same school (e.g. Mr. Jayasinghe's own daughter attends Green Valley).	The system keeps the two relationships on separate, independently scoped logins. His Teacher account never inherits Parent-level access to his daughter's class if he doesn't teach it, and his Parent view of his daughter never grants Teacher-level access to her class.
A substitute teacher covers Grade 10A for two weeks while Mr. Jayasinghe is on leave.	The Administrator grants the substitute a time-boxed Teacher role scoped only to 10A and only for the leave period; access expires automatically at the end date without manual clean-up.
Grade promotion at the end of the academic year (all students move up a grade).	Only the Principal or School Administrator can trigger the bulk promotion process; Teachers cannot promote or move students between classes individually, preventing accidental or unauthorized reclassification.
A parent disputes an attendance mark and wants it corrected.	The Parent role has no edit access; the correction must be made by the Teacher of record (with the change logged in the audit trail) or, if the Teacher is unavailable, by the Administrator on the Teacher's behalf, still logged under the Administrator's identity.
An Education Officer asks a school to explain a specific student's repeated absences.	The conversation happens outside GSMS's aggregated reporting view; if zonal oversight of a named case is ever required, it is handled as a Principal-level export request, not a standing Education Officer permission, keeping individual student data out of routine cross-school access.

16. Audit and Override Discipline

Every action outside a role's normal scope — a Principal correcting a published mark, an Administrator entering data on a teacher's behalf, a time-boxed substitute grant — is written to the append-only audit log defined in the Software Design Specification (GSMS-SDS-001, Section 6), capturing who acted, on whose behalf, what changed, and when. Overrides are a deliberate, logged exception mechanism, not a hidden back door: the goal is that Green Valley's staff can always reconstruct, after the fact, exactly who touched any record and why.